



Sojourn Compass Mobile App

A Business Application Development Proposal Presented to
Department in Information Technology
College of Computer Studies
De La Salle University

In Partial Fulfillment of the Requirements in
ITISDEV: Business Applications Development

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Date Submitted

February 15, 2026

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Proposal Outline

1. Executive Summary

This proposal outlines the development of the Sojourn Compass Mobile Application, a digital ecosystem designed to address critical gaps in coordination, communication, and engagement during organized pilgrimages. The project aims to replace the current fragmented system of rapidly changing and outdated itineraries, verbal instructions, and manual tracking with a centralized, user-friendly mobile platform.

The primary objectives are to unify information dissemination, enhance the spiritual and educational experience through gamification, and establish a real-time feedback loop between pilgrims and coordinators. The expected outcomes include a measurable reduction in coordination errors, improved participant satisfaction and accessibility, and the creation of a unified, historical database for future pilgrimage programs.

2. Background and Context

During the Pilgrimage, there exists a significant engagement and coordination gap. Current operational methods rely on a disjointed collection of tools, including spreadsheets for itineraries, physical handouts, and verbal announcements. This fragmented approach leads to the core problem identified as disorganized on-trip coordination & documentation. Information is often lost, misunderstood, or delivered too late, while participant feedback is delayed, preventing real-time issue resolution.

Furthermore, pilgrims face accessibility & navigation issues, struggling with imprecise meeting point descriptions, language barriers at historical sites, and a lack of a unified source for critical information like packing lists. This application serves as a centralized digital companion, designed to transform the pilgrimage experience. It replaces a menagerie of different platforms with interactive navigation, incentivizes engagement through gamification, and creates a direct communication line between pilgrims and the coordinating team, directly addressing the inefficiencies highlighted in the Ishikawa diagrams.

The core problem is the lack of a unified, accessible, and interactive information system for managing large-group pilgrimages. This results in operational inefficiencies for coordinators and a diminished, often confusing, experience for pilgrims. This project aims to solve this by developing a comprehensive mobile application.

3. SWOT Analysis

Table 1: SWOT Analysis Summary Table

SWOT Analysis	
Strengths	• Coordinators possess deep historical knowledge and a vast network within the Lasallian community
Weaknesses	• Fragmented Information & Lack of a Unified Data Source • Coordination tasks like headcounts and itinerary updates are done manually that wastes time and increases error risk. • Ineffective Feedback Loop
Opportunities	• Demand for Multilingual Access • Mobile technology can be used to create a centralized platform • AR features and digital guides can overcome language barriers
Threats	• Pilgrims, especially the elderly, can get lost in crowded, unfamiliar European cities • Unexpected closures of pilgrimage sites and varying weather conditions can disrupt itineraries • Relying on third-party apps like Viber makes the program vulnerable to their limitations and policy change • The lack of digital literacy among the older pilgrims could hinder adoption of a digital solution

3.1. Strengths

The Sojourn's greatest inherent strength lies in its deeply knowledgeable and dedicated leadership team, whose expertise forms the spiritual and logistical backbone of the pilgrimage. As detailed in the interviews, the team is composed of individuals with specialized roles: a command center coordinator for problem-solving, a spiritual guide for daily religious activities, a local point person for interacting with pilgrims and venues, and most notably, a historian. This historian, Brother Mawel, possesses vast knowledge about European and Lasallian history, speaks multiple languages, and provides invaluable, on-the-ground historical context that transforms simple sightseeing into a rich, educational experience. This depth of human capital and institutional memory is irreplaceable and provides a level of personalized guidance that no commercial tour could offer.

Furthermore, the program benefits from an established and trusted organizational framework within the De La Salle Philippines network. There is a clear, albeit sometimes cumbersome, financial process in place that accommodates participants from various economic backgrounds through options like salary deductions. This institutional backing provides a sense of security and legitimacy, from the accounting office that manages loans to the established relationships with external partners like TLScontact for visa processing. The coordinators also leverage their extensive personal and professional networks, which can lead to spontaneous, enriching experiences that are impossible to pre-plan and add unique value to the pilgrimage. This combination of profound historical expertise and a resilient, well-connected organizational structure ensures that the core mission of the pilgrimage is always fulfilled, providing a stable and trusted foundation upon which a digital tool can be built to enhance, rather than replace, the human elements of the journey.

3.2. Weaknesses

The weakness currently plaguing the Sojourn program is its fragmented and unreliable information system, or lack thereof. Essential data is dispersed across a chaotic mix of outdated Google

sheets, verbal announcements made during busy moments, and scattered Viber and Messenger chats. This lack of a centralized, authoritative source leads to constant confusion, as pilgrims are often left relying on a more organized participant's personal spreadsheet or scrambling to find critical updates buried in long chat histories. The problem is exacerbated by itineraries and schedules that are subject to last-minute, spontaneous changes, yet these adjustments are often communicated only verbally upon exiting the bus, leaving no digital record for participants who may have missed it or for future planning. This disorganization directly contributes to a significant documentation gap, where a treasure trove of photos, reflections, and the lectures from the team's historian are lost because there is no easy, intuitive system for uploading and archiving them.

Compounding the information chaos are inefficient, manually-executed coordination tasks that are both time-consuming and prone to human error. Team leaders, like Professor FKF in the interview, are forced to conduct physical headcounts to track attendance and rely on memory or separate apps to identify members they've just met. The need for a streamlined attendance and identification system is clear. Furthermore, the current feedback mechanism is largely ineffective, typically consisting of a single, lengthy form distributed after the pilgrimage has ended. By that point, participants are tired and have forgotten the specific details of their experiences from weeks prior, resulting in low response rates and low-quality data that offers no opportunity for real-time issue resolution during the trip itself.

Finally, a major vulnerability lies in the program's handling of sensitive documentation and the persistent issue of forgotten items. The practice of using a public, unsecured Google Drive for everyone to upload personal documents like bank statements and passports represents a severe data privacy violation that exposes all participants to significant risk. On a more logistical but equally stressful level, pilgrims frequently forget essential items, most notably the "Pilgrim's Passport" booklet used for collecting stamps. The lack of a

pre-departure or daily checklist system means that participants, especially the elderly who make up the majority of the group, struggle to keep track of what to bring each day, from specific religious items like candles and clam shells to appropriate clothing for unpredictable weather, leading to last-minute stress and missed opportunities for engagement.

3.3. Opportunities

The opportunities for the Sojourn program lies in harnessing mobile technology to create a centralized, user-friendly application that directly addresses its current fragmentation. By consolidating itineraries, announcements, historical content, and documentation into a single, secure application, the program can eliminate the chaos of scattered Google sheets, unsecured public drives, and missed verbal instructions. This platform could serve as the official source of announcements, with real-time updates pushed directly to every pilgrim's phone, ensuring that everyone from the 33-year-old to the 81-year-old participant is working from the same, current information.

Furthermore, technology presents a powerful opportunity to dramatically enhance the spiritual and educational engagement of the pilgrims, turning a passive tour into an interactive journey. The interviews highlighted the deep desire for better documentation and reflection, with pilgrims like Professor FKF expressing a wish to attach notes and voice recordings to their photos to preserve the context and emotions of the moment. An application could satisfy this through an integrated digital reflection journal, allowing users to log thoughts, record audio of the historian's lectures, and pair media with specific GPS locations. This same technology could be used to gamify the experience through a digital badge system, appealing to the natural inclination for collection seen in the physical "Pilgrim's Passport." By creating a digital scrapbook or a bucket list of sites and reflections with rewards for completion, the program could boost morale and provide a lasting, meaningful digital keepsake long after the physical journey ends.

The application can leverage its data collection capabilities to generate powerful insights. A real-time feedback loop, with short, periodic check-ins every few days rather than a single post-trip survey, would provide coordinators with immediate, actionable data. They could see which sites resonated most, identify logistical pain points as they happen, and gauge participant satisfaction in the moment. This data, anonymized and aggregated, could then inform future itinerary planning, helping coordinators make data-driven decisions about which hotels, venues, and activities that provide the highest value, iterating on the pilgrimage during the tour.

Finally, the unique challenges of the demographic present an opportunity to design an inclusive and accessible solution. Features like an intuitive interface with options for larger text, high-contrast modes, and clear icons would ensure that the oldest participants are not left behind. The integration of GPS-based team tracking, building on the successful pilot by Professor FKF's group, addresses a core safety concern. For the elderly participants who may wander or have difficulty keeping pace, or for anyone who gets lost in crowded European cities, this feature provides peace of mind for both the individual and the group leaders.

3.4. Threats

The successful implementation and adoption of a digital solution for the Sojourn program face significant external threats, foremost among them being the inherent technological and demographic challenges posed by the participant base. As the interview data reveals, the pilgrimage demographic skews considerably older, with participants ranging from their late 40s to over 80 years of age. This presents a risk regarding digital literacy and user adoption. Elderly users may exhibit resistance to adopting a new, unfamiliar technological platform, particularly if the interface is not designed with accessibility considerations such as intuitive navigation, adjustable text sizing, and high-contrast visual elements. If a critical mass of participants cannot or will not utilize the

application effectively, the purported benefits of centralized coordination and real-time communication are negated, potentially forcing coordinators to maintain parallel analog systems and thereby defeating the purpose of digital transformation.

Environmental and situational volatility during the pilgrimage itself poses another substantial threat to the application's reliability and utility. The interviews documented numerous unanticipated disruptions, including sudden public transportation problems in Paris, unexpected closures of religious sites due to renovations or events, and weather conditions affecting daily plans. These variables, entirely outside the control of the program coordinators, directly impact the accuracy of any pre-planned digital itinerary. If the application cannot accommodate and disseminate real-time updates regarding such disruptions instantaneously, it risks becoming a source of misinformation rather than a reliable coordination tool. Furthermore, the very nature of the pilgrimage introduces safety and security threats. Participants, particularly elderly members, are vulnerable to getting lost in unfamiliar cities with complex public transportation systems and potential language barriers. While the application may offer GPS tracking, this feature itself could give a false sense of security or, conversely, face rejection from privacy-conscious participants, as evidenced by the resistance encountered during Professor FKF's Life360 proposal by other teams.

The dependency on third-party platforms and infrastructure presents a critical external threat to the integrity and sustainability of the proposed digital ecosystem. The current program's reliance on applications like Viber and Messenger exposes it to the limitations and policy changes of external vendors, such as the inability to pin multiple announcements simultaneously. A custom application, while mitigating some of these specific constraints, becomes similarly dependent on the continued availability and affordability of cloud service providers, mapping APIs (Google Maps or Apple Maps), and mobile operating system compatibility. Changes in API pricing structures, deprecation of key features, or shifts in data privacy

regulations could impose unforeseen operational costs or require technical reconfiguration.

Finally, a threat lies in the unpredictable nature of human behavior and sustained engagement. The interviews consistently highlighted the phenomenon of enthusiastic participation during the pilgrimage giving way to post-trip disengagement, exemplified by the empty shared photo drive and the negligible response rate to end-of-journey feedback forms. This behavioral pattern threatens the long-term viability of features reliant on ongoing user contribution, such as digital scrapbooks, reflection journals, and post-pilgrimage feedback mechanisms. If participants, overwhelmed by the intensity of the journey or preoccupied with resuming professional responsibilities, fail to upload content or complete evaluations, the application fails to capture the collective memory of the experience and loses the opportunity to generate actionable insights for program improvement. This threat extends to the gamification elements as well; while digital badges may appeal to certain demographics, elderly participants accustomed to tangible keepsakes like physical stamps may find virtual rewards unengaging or immaterial, undermining the motivational objectives of such features. Without careful attention to user psychology and sustained incentive structures, the application risks becoming abandoned by its own user base.

4. Ishikawa Diagram

4.1. Accessibility & Navigation

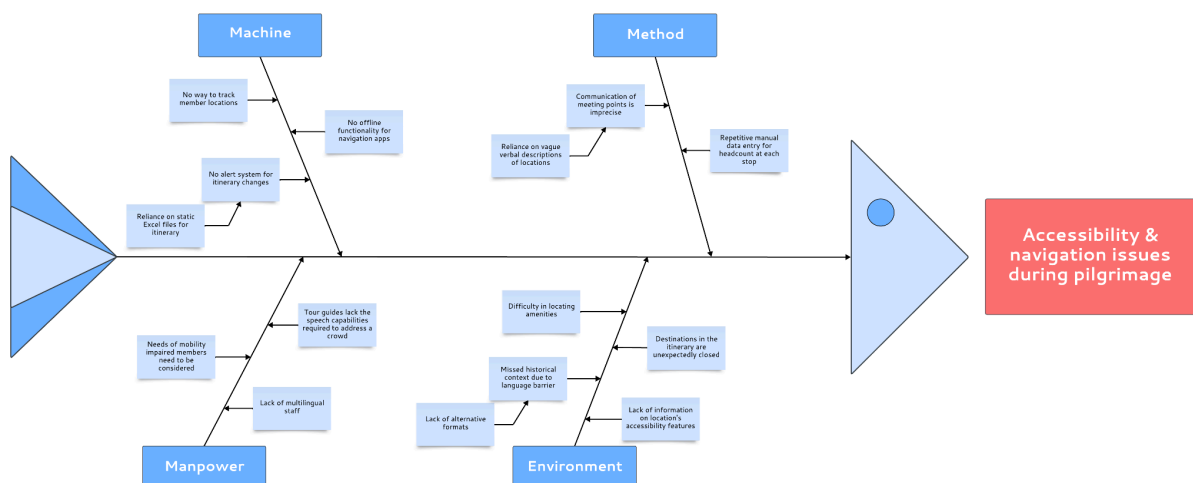


Figure 1: Accessibility & Navigation Issues during Pilgrimage Diagram

The diagram identifies "Disorganized On-Trip Coordination & Documentation" as the central problem, with root causes categorized under Manpower and Method. Under Manpower, pilgrims frequently forget essential items like the Pilgrim's Passport and religious articles due to the absence of systematic daily reminders, while inconsistent documentation practices. Digital literacy gaps among elderly participants, combined with resistance to purchasing mobile data, create communication blind spots, leaving some individuals unreachable for real-time updates. Additionally, the tour historian's declining audibility leads to missed historical context, and participants lack proper guides for navigating processes like foreign transportation systems, exacerbating confusion during free time.

Under Method, the absence of a structured feedback loop prevents real-time issue resolution, with coordinators relying on a single, ineffective post-trip survey that yields poor response rates and low-quality data. Critical itinerary changes are communicated only through verbal announcements the night before or upon bus departure, leaving no digital record for participants who may have missed them. Official communication channels like Viber are improperly utilized, with important messages buried in chat histories due to the platform's inability to pin multiple announcements simultaneously. These methodological failures are compounded by external factors that the current fragmented system cannot adapt to efficiently.

4.2. Disorganized On-Trip Coordination & Documentation

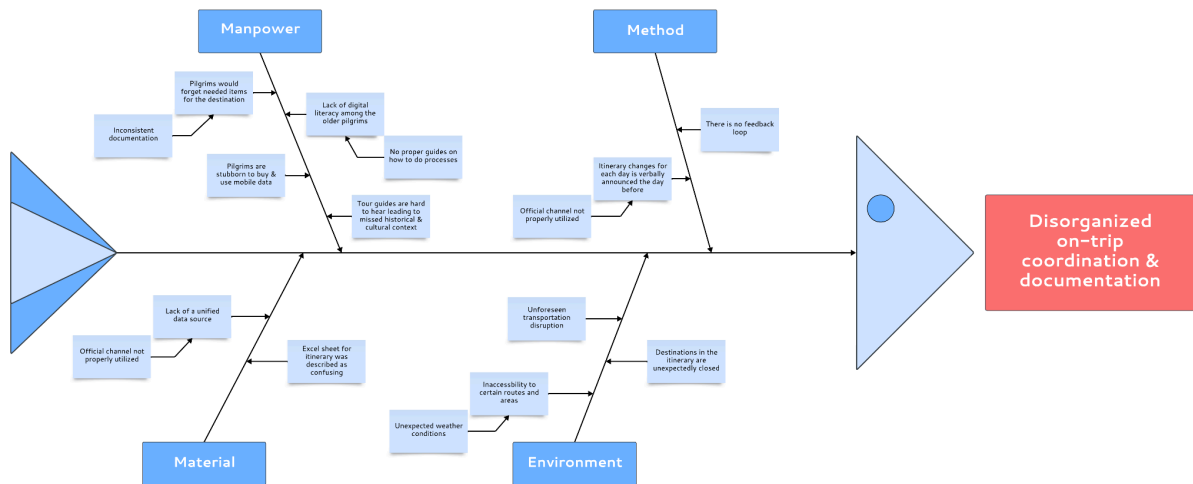


Figure 2: Disorganized On-Trip Coordination & Documentation Diagram

This diagram identifies "Accessibility & Navigation Issues During Pilgrimage" as the central problem, with causes primarily through the Machine category while Method factors are implied throughout. The analysis reveals that pilgrims face persistent barriers to independent movement and information access due to systemic technological gaps and infrastructural inadequacies.

The diagram highlights how the absence of real-time tracking, offline navigation capabilities, and automated alert systems leaves participants vulnerable to getting lost, missing itinerary changes, and struggling to locate essential amenities in unfamiliar European cities. Communication breakdowns stemming from language barriers and imprecise meeting point descriptions compound these navigation challenges. Accessibility considerations for mobility-impaired members are entirely unaddressed, with no solid advance information about site accessibility features or alternative routes. The continued reliance on static Google sheets and manual processes for coordination ensures that accessibility remains an afterthought rather than an integral design consideration, fundamentally compromising the pilgrimage experience for participants across all mobility levels and language proficiencies.

5. Solutions Framework

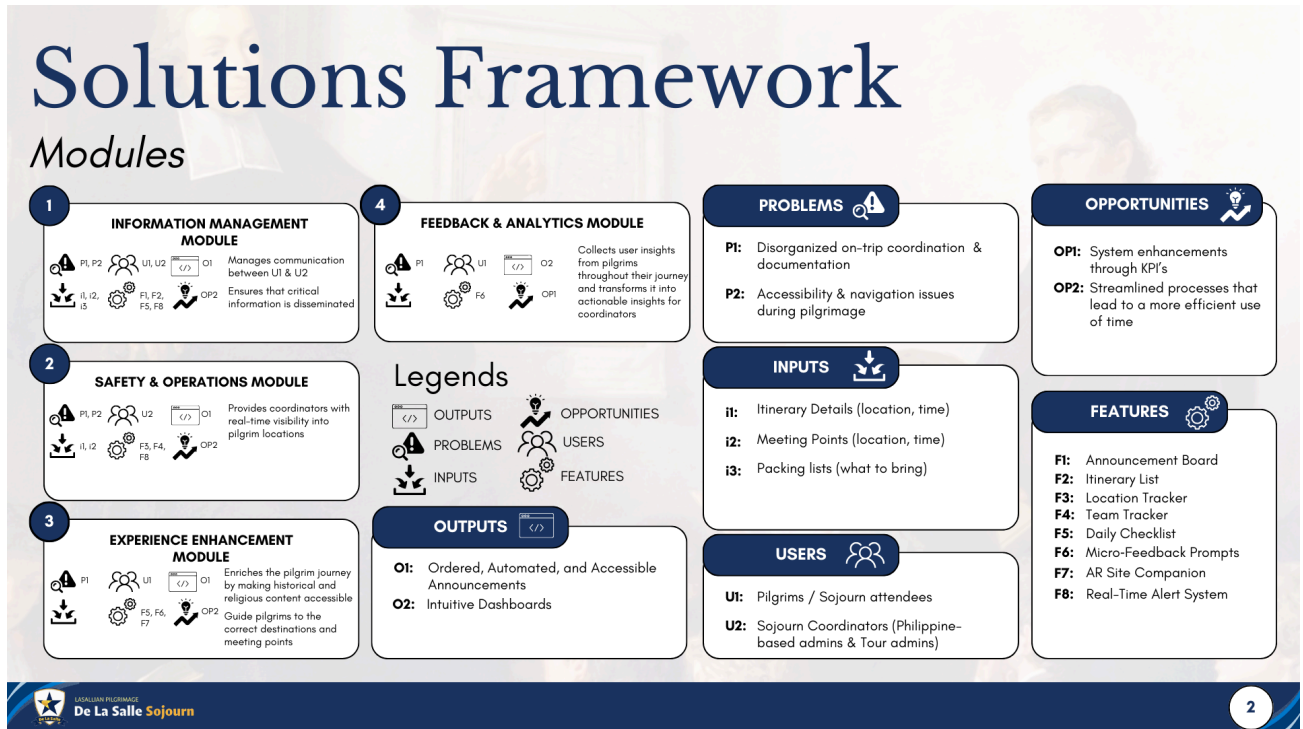


Figure 3: Solutions Framework

The proposed Solutions Framework for the Sojourn Compass Mobile Application is structured around four interconnected functional modules, each designed to address specific documented problems while capitalizing on identified opportunities for program enhancement. This ensures systematic coverage of the pilgrimage ecosystem, from pre-departure preparation through on-trip coordination to post-journey analysis, while maintaining clear separation of responsibilities between system components. The framework operates on a defined input-output model, consuming critical data elements including itinerary details, meeting point specifications, and packing requirements, and transforming these through feature-rich modules into tangible outputs comprising ordered, accessible announcements and intuitive user dashboards. The architecture explicitly acknowledges the dual-user nature of the system, accommodating both pilgrims attending the journey and coordinators managing operations from the Philippines and on-site in Europe, with each module implementing appropriate access controls and interface variations based on user role requirements.

5.1. Information Management Module

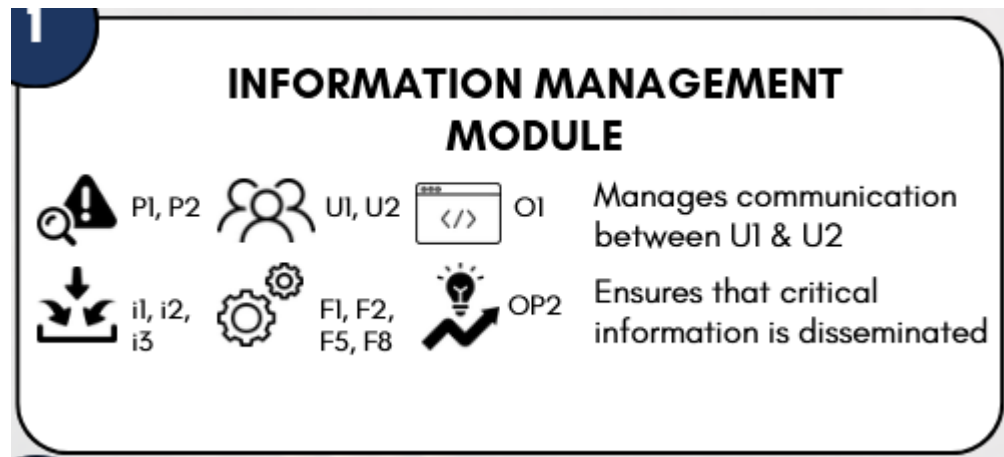


Figure 4: Information Management Module

The Information Management Module serves as the centralized nervous system of the application, addressing the core problem of disorganized coordination and fragmented documentation. It consumes raw input data including itinerary details with corresponding locations and times, meeting point specifications, and comprehensive packing lists, then processes these through a robust communication layer before disseminating refined information to both pilgrim and coordinator users. The module powers the Announcement Board, which replaces scattered Viber and Messenger chats with a centralized repository where coordinators can pin multiple critical updates simultaneously, eliminating the problem of important messages becoming buried in conversation threads. The Itinerary List feature transforms confusing, outdated Google sheets into structured, visually intuitive daily schedules enriched with site photographs and historical context. When coordinators make schedule changes or add spontaneous destinations, updates propagate instantaneously to all connected devices, eliminating the confusion caused by verbal-only announcements.

5.2. Safety and Operations Module

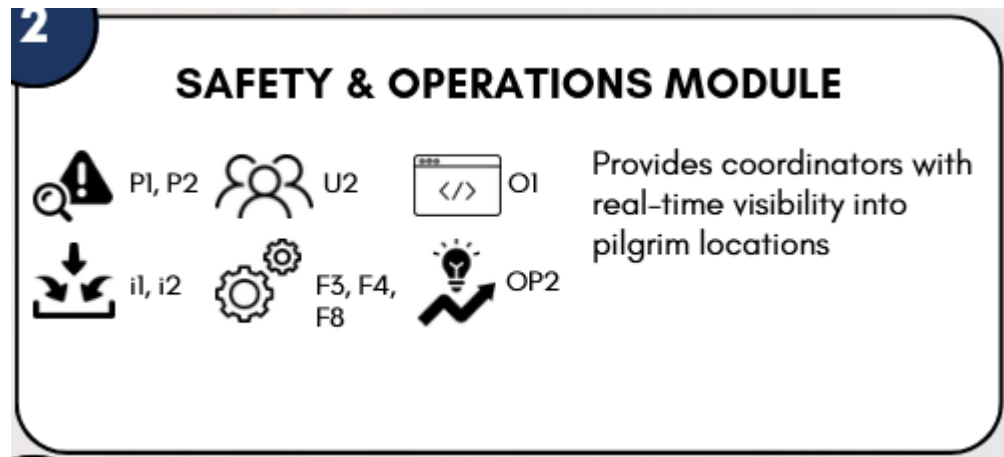


Figure 5: Safety & Operations Module

The Safety and Operations Module addresses the intersection of disorganized coordination and accessibility issues by providing real-time visibility into pilgrim locations and establishing reliable emergency communication channels. This module directly responds to documented safety threats involving elderly participants navigating unfamiliar European cities, language barriers during transportation disruptions, and the logistical challenges of manual headcounts before each bus departure. Feature implementation encompasses the Location Tracker, providing individual pilgrims with turn-by-turn navigation assistance to meeting points, while the Team Tracker enables leaders to visualize all assigned members on a unified map interface and receive automated proximity alerts when individuals deviate from designated areas. The Real-Time Alert System establishes an emergency communication channel capable of broadcasting urgent notifications, such as transportation strikes, site closures, or security incidents, to all connected devices with configurable priority levels. The module automates attendance verification through GPS proximity detection, generating real-time absence reports that enable coordinators to identify missing individuals before departure.

5.3. Experience Enhancement Module

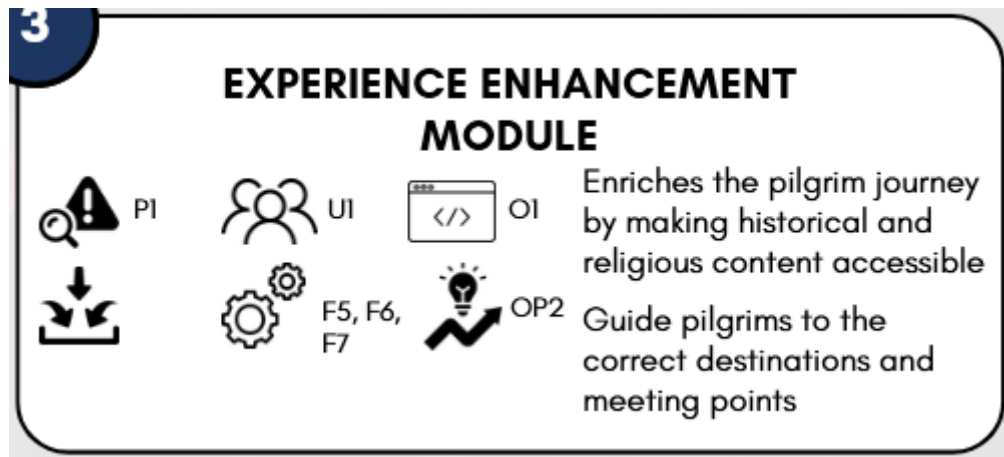


Figure 6: Experience Enhancement Module

The Experience Enhancement Module focuses on enriching the pilgrim journey by making historical and religious content accessible while providing intuitive guidance to correct destinations and meeting points throughout the European itinerary. This module directly addresses accessibility problems where participants struggle to appreciate sites due to language barriers, limited historical context, and the ephemeral nature of verbal lectures delivered by aging historians. The Daily Checklist feature synthesizes itinerary requirements, weather forecasts, and Sojourn-specific luggage regulations into personalized packing recommendations delivered each evening, directly addressing the recurring problem of forgotten items like the Pilgrim's Passport and religious articles. The AR Site Companion applies augmented reality technology to overlay historical information onto physical environments, enabling pilgrims to point device cameras at landmarks and receive contextual narratives and translated descriptions.

5.4. Feedback and Analytics Module

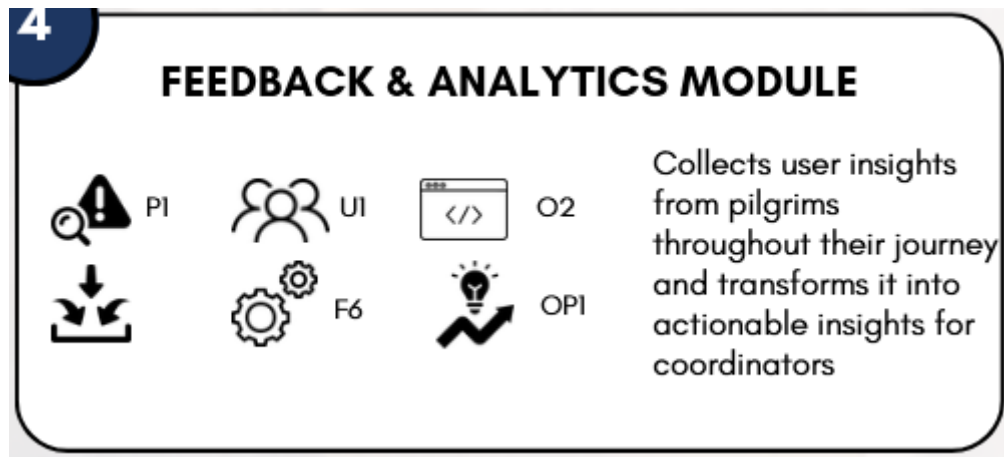


Figure 7: Feedback & Analytics Module

The Feedback and Analytics Module functions as the intelligence gathering and processing layer of the application ecosystem, systematically collecting user insights throughout the pilgrimage journey and transforming raw experiential data into actionable strategic intelligence for program coordinators. This module directly addresses the documented ineffectiveness of current feedback mechanisms, where single post-trip surveys yield minimal response rates and poor data quality due to participant memory decay. The Micro-Feedback system delivers brief, targeted surveys at strategically determined intervals with each prompt requiring minimal completion time to maximize response rates. The collected data streams into a centralized analytics engine applying aggregation algorithms and pattern recognition to identify emerging trends requiring coordinator attention. Coordinators gain visibility into satisfaction trajectories throughout the journey, enabling real-time interventions when negative patterns emerge. Historical data accumulation enables predictive analytics, allowing administrators to forecast which itinerary elements resonate with specific demographic segments and optimize vendor selection based on empirical satisfaction data rather than intuition.

6. Scope of Work

The Sojourn Compass application will be built around the following core functions, accessible after a secure login process as outlined in the preliminary BPMN diagram.

Table 1: Scope of Work Summary Table

Function	Description
Log-in	The user must log-in before accessing the other features of the app. This requires the user to enter his/her registered email.
Live Itinerary with Real-Time Updates	A card-based daily itinerary displaying site names, photos, descriptions, and scheduled times.
Centralized Announcement Board	A dedicated screen displaying multiple pinned messages simultaneously. Messages appear in reverse chronological order with pinned items remaining at the top. This board retains all important messages without requiring scrolling through chat history.
Team Tracker	A map-based view showing the real-time locations of the user's team members (for leaders) or leaders/coordinators (for members), accessible only during the active pilgrimage window.
Daily Smart Checklist	A personalized packing checklist based on: day's itinerary (clam shell, pilgrim passport requirements), current weather forecast (temperature, rain probability), and Sojourn-specific luggage rules (bus vs. flight limits).
Activity Guide Repository	A digital library that provides instructions for tasks, historical context for landmarks, and safety protocols, all accessible offline.

Reflection Journal	A dedicated space for participants to log thoughts, upload photos, or answer prompts. This would transform a walk into a meaningful learning experience.
Digital Badge System	A gamification layer where users unlock rewards for completing routes, finishing reflections, or visiting all checkpoints. This boosts retention and morale.
Real-Time Feedback Loop	A direct portal where participants can report issues or ask questions. This data is instantly pushed to the coordinating team's dashboard.
Secure Document Vault (Resource Digitization)	A "Vault" feature that hosts all program materials (PDFs, videos, schedules), ensuring participants have everything they need in their pocket.

6.1. Log-in Function

The authentication feature serves as the secure gateway to the app, implementing a system that ensures data privacy and personalized user experiences. Upon launching the application, users are required to authenticate using their registered institutional email addresses, using Google-based authentication protocols to eliminate the need for managing separate credentials while benefiting from established security infrastructure. The system maintains persistent session management to accommodate users with varying levels of technical proficiency, remembering authenticated devices to minimize repetitive login requirements while implementing automatic timeout protocols for security compliance. This granular approach to access control directly addresses the data privacy vulnerabilities identified in current operations, ensuring that sensitive documents and personal location data remain visible only to authorized individuals throughout the pilgrimage journey.

6.2. Live Itinerary with Real-Time Updates

The dynamic itinerary feature revolutionizes the current fragmented scheduling system by providing pilgrims with a centralized, continuously updated digital representation of their daily pilgrimage activities. This feature presents information through an intuitive, card-based interface where each day's activities are displayed with corresponding site names, high-quality photographs, descriptive historical context, and precisely scheduled timings. Unlike the static, rapidly obsolete Google sheets currently employed, this module maintains persistent synchronization with the coordinating team's master schedule, automatically pushing updates to all connected devices the moment changes are approved. When unexpected opportunities arise or when unforeseen closures necessitate itinerary adjustments, coordinators can modify the digital schedule instantaneously, eliminating the confusion caused by verbal announcements made upon exiting buses. The interface supports multiple viewing modalities, allowing users to examine their entire pilgrimage trajectory chronologically or drill down into individual days for detailed activity breakdowns. Each itinerary card integrates seamlessly with the mapping module, providing one-tap navigation to meeting points and site locations, while historical photographs and contextual descriptions enrich the user's understanding before physical arrival at each destination.

6.3. Centralized Announcement Board

The announcement board feature functions as the authoritative, single-source communication channel that eliminates the current dependency on scattered Viber and Messenger conversations where critical information becomes irretrievably buried. This dedicated screen implements a sophisticated yet accessible message management system that displays multiple concurrently pinned announcements, ensuring that urgent logistical updates, prayer schedule changes, and safety advisories remain permanently visible at the top of the interface regardless of subsequent message volume. Messages appear in reverse chronological order, with the system automatically archiving older announcements while

maintaining complete searchability for future reference. Administrators possess granular control over message prioritization, categorization, and expiration, enabling them to designate certain communications as persistent while allowing routine updates to cycle through naturally. This centralized approach addresses the fundamental limitation of existing platforms where only a single message can be pinned, forcing team leaders like Professor FKF to repeatedly scroll through chat histories to retrieve important instructions. Furthermore, the board supports multimedia attachments, allowing coordinators to supplement text announcements with relevant images, PDF documents, or short video explanations, thereby accommodating diverse learning preferences among the elderly demographic while ensuring no participant misses critical information due to communication fragmentation.

6.4. Team Tracker

The team tracking feature implements a privacy-conscious, GPS-based location visualization system designed to enhance participant safety and streamline group coordination throughout the pilgrimage. This feature presents a map-based interface displaying the real-time geographical positions of team members relative to one another, accessible exclusively during the active pilgrimage window to address privacy concerns that previously derailed similar initiatives like the Life360 proposal. For team leaders, the interface reveals the locations of all assigned members, enabling efficient headcount verification before bus departures and rapid identification of individuals who may have wandered into unfamiliar urban environments. Standard pilgrims, conversely, can view the positions of their designated leaders and coordinators, providing autonomous navigation capability when exploring during free time while maintaining confidence in their ability to reunite with the group. The module integrates with the itinerary system to display designated meeting points directly on the map, with automated proximity alerts notifying users when they deviate significantly from planned gathering locations. This functionality directly addresses documented safety threats involving elderly participants navigating complex European

cities with limited language proficiency, while the time-limited activation window mitigates resistance from privacy-conscious users who objected to continuous, indefinite location monitoring. The system operates efficiently even under variable connectivity conditions, caching location data locally and synchronizing when network access becomes available.

6.5. Daily Smart Checklist

The checklist feature represents integration of itinerary data, real-time environmental information, and program-specific logistical requirements to generate personalized, contextually relevant packing recommendations for each pilgrimage day. Rather than presenting a static, one-size-fits-all inventory (or none at all), this feature dynamically compiles checklists by analyzing three distinct data streams: the day's scheduled activities (identifying requirements for items such as clam shells for specific blessings or the Pilgrim's Passport for stamp collection), current and forecasted weather conditions (recommending umbrella carriage during predicted precipitation or jacket requirements for alpine excursions), and Sojourn-specific luggage regulations (distinguishing between items permitted on buses versus those restricted to checked luggage). The interface presents these recommendations through an intuitive, checkable interface that allows users to verify item preparation visually, with the system maintaining historical records of checklist completion to identify patterns in forgotten items for future program improvement. This functionality directly addresses the recurring problem of participants, particularly elderly members, forgetting essential items like the Pilgrim's Passport or failing to prepare appropriate attire for unpredictable weather conditions. By shifting item preparation from memory-dependent recall to systematic, data-driven guidance delivered the evening before each day's activities, the module reduces pre-departure stress and ensures all pilgrims arrive at meeting points fully equipped for the day's spiritual and logistical requirements.

6.6. Activity Guide Repository

The activity guide repository functions as a comprehensive, offline-accessible digital library that consolidates all instructional, historical, and safety-related content into a single, searchable knowledge base available to pilgrims throughout their journey. This feature addresses the critical documentation gap identified in the interviews, where the invaluable historical lectures delivered by Brother Mawel currently exist only as ephemeral verbal communications, vulnerable to loss and inaccessible to future pilgrimage cohorts. The repository hosts structured content organized by geographical location, historical period, and thematic relevance, enabling users to access detailed narratives about sites before, during, or after physical visitation. Each entry supports multiple media formats, including text, high-resolution images, audio recordings of lectures, and short explanatory videos, accommodating various learning preferences while ensuring accessibility even in connectivity-constrained environments through intelligent offline caching. The system implements robust search functionality, allowing pilgrims to query by location name, historical figure, or keyword, transforming the repository from passive content storage into an active educational tool. Furthermore, the module includes practical instructional content addressing logistical challenges documented in the interviews, including illustrated guides to navigating European public transportation systems, explanations of metro card purchasing procedures at shops, and phrasebook-style language assistance for common French and Italian interactions. This comprehensive approach ensures that every participant, regardless of prior international travel experience, possesses the knowledge required to navigate unfamiliar environments confidently and safely.

6.7. Reflection Journal

The reflection journal feature provides pilgrims with a structured, multimedia-enabled digital space for documenting personal spiritual experiences, recording historical insights, and preserving the emotional context of their pilgrimage journey. This feature directly addresses Professor FKF's documented frustration

with accumulating thousands of photographs while losing the associated thoughts, feelings, and historical contexts that rendered those images meaningful. The journal interface supports multiple input modalities, accommodating diverse user preferences through traditional text entry, voice recording with automated speech-to-text transcription, and direct camera integration that automatically couples images with GPS location data and user-generated annotations. Each journal entry can be associated with specific itinerary stops, enabling pilgrims to build a comprehensive, geo-tagged narrative of their spiritual journey that chronologically maps to their physical movements throughout Europe. The module implements sophisticated offline functionality, allowing users to capture thoughts and media during bus journeys or site visits without active internet connectivity, with automatic synchronization occurring when network access becomes available. This capability proved particularly valuable during the interviews, where Professor FKF described surreptitiously recording Brother Mawel's historical lectures during bus rides, transforming passive travel time into productive documentation sessions. The journal further supports optional sharing capabilities, enabling pilgrims who wish to exchange reflections with family members or fellow participants while maintaining granular privacy controls for those who prefer personal, introspective documentation. By transforming fleeting thoughts into permanent, searchable records, this module ensures that the profound spiritual and educational experiences of the pilgrimage remain accessible for personal reflection long after physical return to the Philippines.

6.8. Digital Badge System

The digital badge system implements a sophisticated gamification layer designed to enhance participant engagement, encourage comprehensive site visitation, and transform the pilgrimage from passive observation into active, rewarding exploration. This feature directly extends the physical "Pilgrim's Passport" concept documented in the interviews, where participants collect stamps from visited churches as tangible souvenirs of their journey. The digital

interpretation maintains this motivational psychology while eliminating the practical problem of forgotten passports and expanding the reward framework beyond simple stamp collection. The system defines multiple badge categories based on diverse achievement criteria, including geographical completion (visiting all sites within a specific region), thematic exploration (engaging with content related to particular saints or historical periods), reflective depth (completing journal entries for designated locations), and social participation (contributing photographs to shared collections). Each badge is associated with specific unlock logic evaluated automatically through system data, including GPS-verified site visits, journal submission timestamps, and content contribution metrics. The interface presents a visually engaging badge collection dashboard where pilgrims can track their progress, discover remaining achievements, and review detailed descriptions of earned recognitions. Individual badge detail pages provide comprehensive information about unlock criteria, associated historical context, and suggestions for related badges that users may find engaging. The system addresses potential demographic concerns regarding digital versus physical rewards by supporting both paradigms. This hybrid approach accommodates elderly users who value tangible keepsakes while leveraging digital engagement to motivate comprehensive participation throughout the journey.

6.9. Real-Time Feedback Loop

The real-time feedback module establishes a direct, structured communication channel between pilgrims and the coordinating team, enabling immediate issue reporting, satisfaction monitoring, and continuous program improvement throughout the active pilgrimage window. This feature directly addresses the documented ineffectiveness of current feedback mechanisms, where a single post-trip survey yields low response rates and poor data quality due to participants' faded memories and post-journey exhaustion. Rather than relying on retrospective evaluation, the module implements periodic, contextually triggered micro-surveys delivered at strategic intervals. These targeted feedback requests present concise, specific

questions about recently completed activities, capturing accurate impressions while the experiences remain fresh in participants' consciousness. The interface supports multiple response formats, including Likert-scale ratings for quantitative analysis, free-text comments for qualitative insights, and optional photograph attachments for visual documentation of issues or highlights. All submissions are instantly transmitted to the coordinating team's administrative dashboard, where designated personnel can monitor emerging trends, identify participants requiring immediate assistance, and generate real-time interventions before minor frustrations escalate into significant dissatisfaction. Historical feedback data is systematically archived and analyzable across multiple pilgrimage cycles, providing the administrative team with actionable insights for long-term itinerary refinement, vendor selection optimization, and participant experience enhancement.

6.10. Secure Document Vault

The secure document vault addresses the critical data privacy vulnerability identified in current operations by providing a protected, access-controlled digital repository for all pilgrimage-related documentation and program materials. This feature directly replaces the egregious practice of uploading sensitive personal documents to publicly accessible Google Drives where any participant could view others' confidential information, a situation that prompted justified alarm from privacy-conscious individuals like Professor FKF during the interviews. The interface presents an intuitive, folder-based organizational structure that categorizes documents by type, required usage date, and processing status, enabling users to quickly verify compliance with visa application requirements and other pre-departure formalities. Administrators can create checklist-driven upload requirements, prompting participants to submit specific documents according to a predetermined timeline while automatically verifying receipt and flagging missing items for follow-up. Beyond personal documentation, the vault serves as the centralized repository for all program materials, including digital copies of the Pilgrim's Passport, historical lecture transcripts, prayer schedules, and

logistical guides, ensuring that every participant carries a complete program library in their pocket throughout the journey. The module implements granular permission controls, allowing document owners to grant temporary, purpose-limited access to third parties such as visa processing personnel or medical providers while maintaining complete revocation authority. This comprehensive approach to document management transforms what was previously a significant data privacy liability into a competitive advantage, demonstrating institutional commitment to participant confidentiality while simultaneously improving operational efficiency through centralized, searchable document access.

7. Gantt Chart

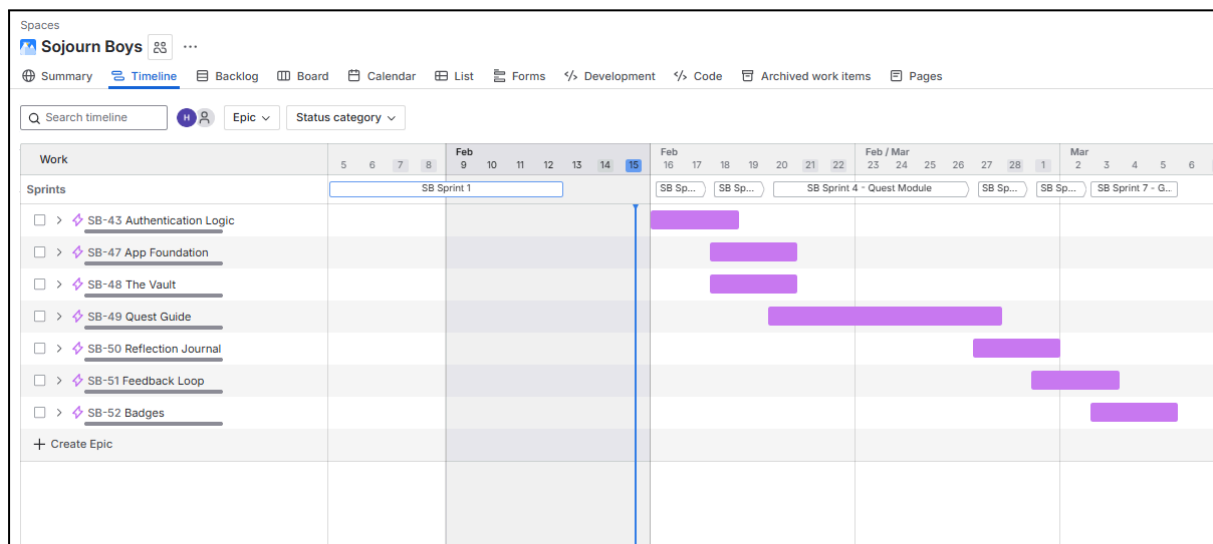


Figure 8: Gantt Chart of Epics And Sprints

Epic	Description (Issues to Create)
Authentication (Sprint 2)	Covers login and access control. Build Google-based email login on the backend, implement role-based access logic (access control matrix), create the frontend login page, and prepare the codebase for role-based UI rendering across the app.

Codebase Scaffolding (Sprint 3)	Sets up core structure and shared views. Build the profile page backend, create separate admin and sojourner dashboards on the frontend, and develop a centralized announcement board with both a frontend view and backend admin CRUD.
Cloud Integration (Sprint 3)	A cloud-backed resource repository. Build the frontend file system view, connect the vault to cloud storage on the backend, and implement scoped media access (PDFs, videos, images) based on user permissions.
Map Module (Sprint 4)	The pilgrimage/quest experience module. Frontend views for the quest map with pin markers, teammate location tracking, timeline list, detail screen (instructions, safety, advisories, notes), and a checklist. Backend admin CRUD for map markers, timeline, quest details, and checklist. System-level integration for team tracking during the pilgrimage window and offline caching of quest data.
Reflection Journal (Sprint 5)	Enables participants to document their experience. Frontend views for journal entry list, detail page, and edit page. Backend support for media attachments, offline editing without WiFi, and logic to couple journal entries with map locations and quests.
Feedback Loop (Sprint 6)	Post-experience feedback collection. Frontend views for the feedback list, detail page, and edit page. Backend CRUD for both sojourner-submitted feedback and admin-side feedback management.
Badges (Sprint 7)	Gamification through achievements. Define badge categories and unlock logic on the backend, build a badge collection dashboard on the frontend, and create a badge detail page/modal for viewing individual badges.

8. BPMN

8.1. AR Process

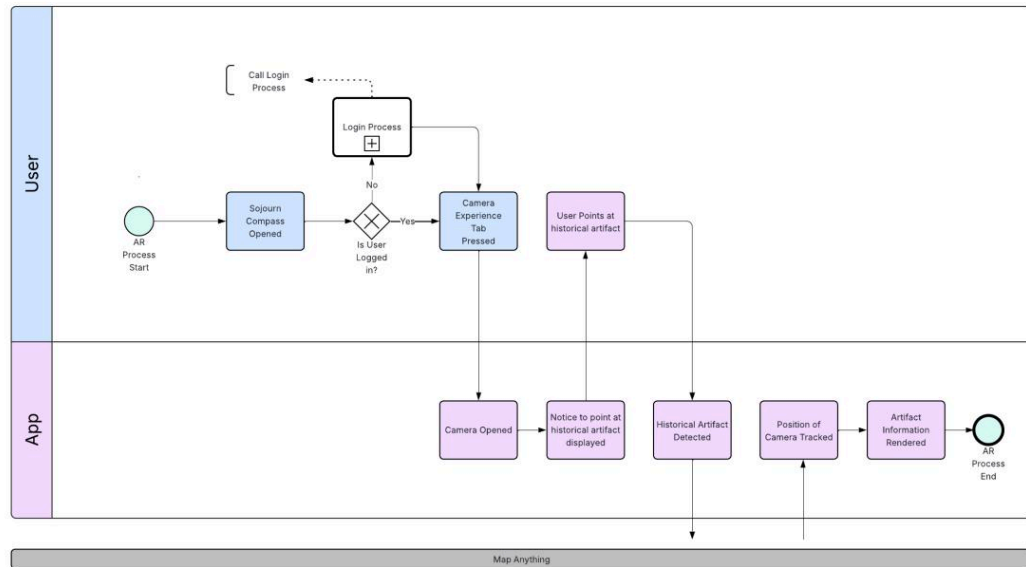


Figure 9: BPMN for AR Process

The Augmented Reality process begins when a user opens the Sojourn Compass application and initiates the camera experience tab. The system first performs a conditional login verification: if the user is not authenticated, the login process is invoked before proceeding; if already logged in, the user proceeds directly to the AR interface. Once authenticated, the application displays a notice instructing the user to point the device camera at a historical artifact. The system continuously analyzes the camera feed until a recognized artifact is detected, at which point it tracks the camera's position relative to the object and renders relevant historical information, contextual narratives, or translated descriptions directly onto the screen overlay.

8.2. Announcement Retrieval Process

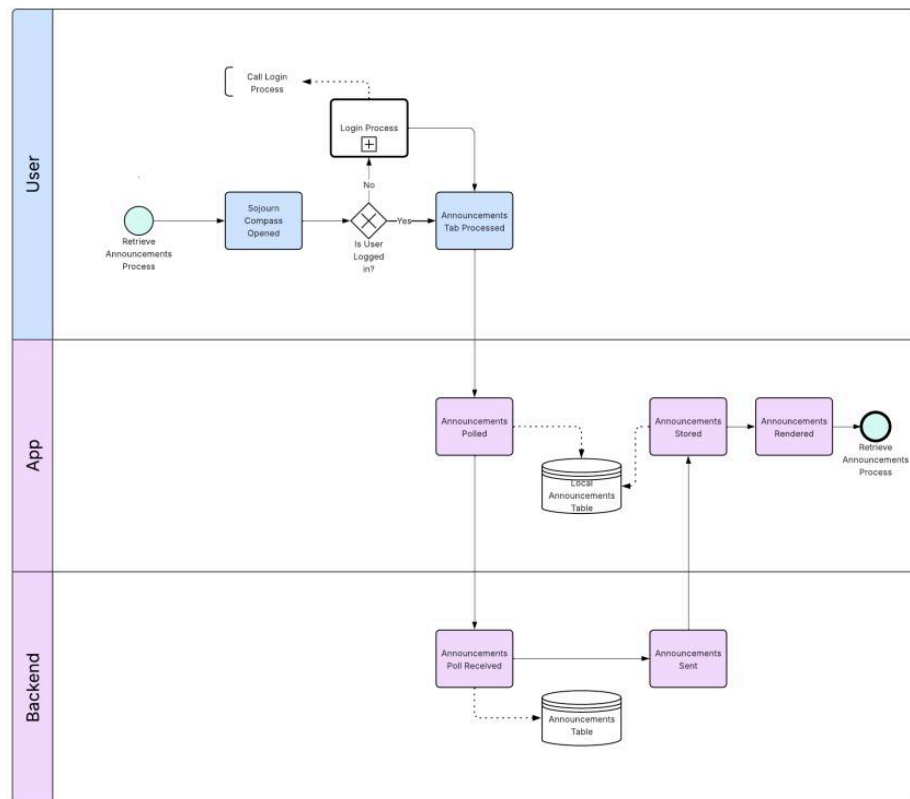


Figure 10: BPMN for Announcement Retrieval Process

The Announcement Retrieval process begins when a user opens the Sojourn Compass application and navigates to the announcements tab. The system first performs a conditional login verification: if the user is not authenticated, the login process is invoked before proceeding; if already logged in, the user gains immediate access to the announcements interface. Upon tab selection, the application initiates a polling request to the backend server, which receives the announcement query and responds with the latest published messages. The application then stores these announcements in a local database table for offline accessibility before rendering them on the user's screen. This dual-layer approach ensures that pilgrims can access critical updates even when cellular connectivity is unreliable during bus travel or in remote areas, directly addressing the documented problem of missed verbal announcements and fragmented communication across scattered chat platforms.

8.3. Login Process

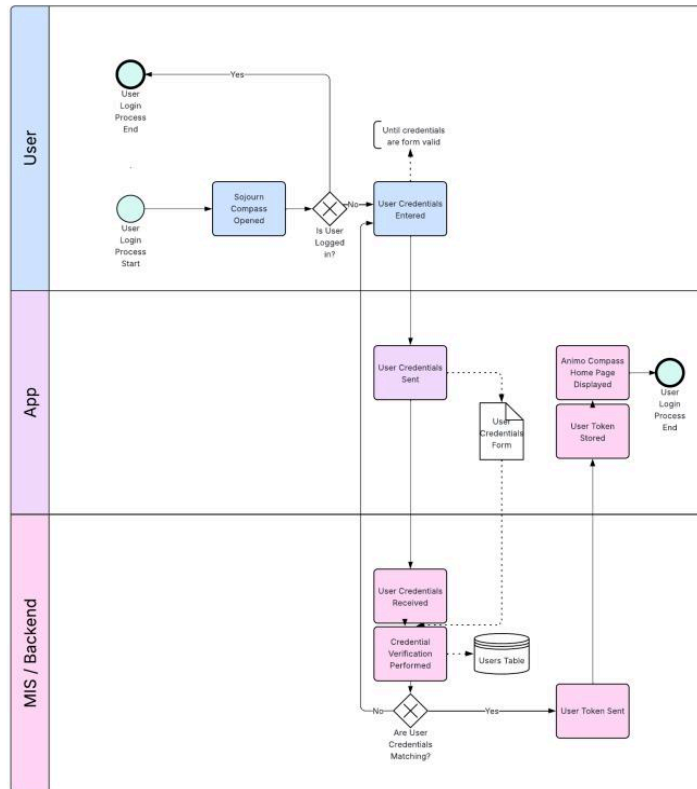
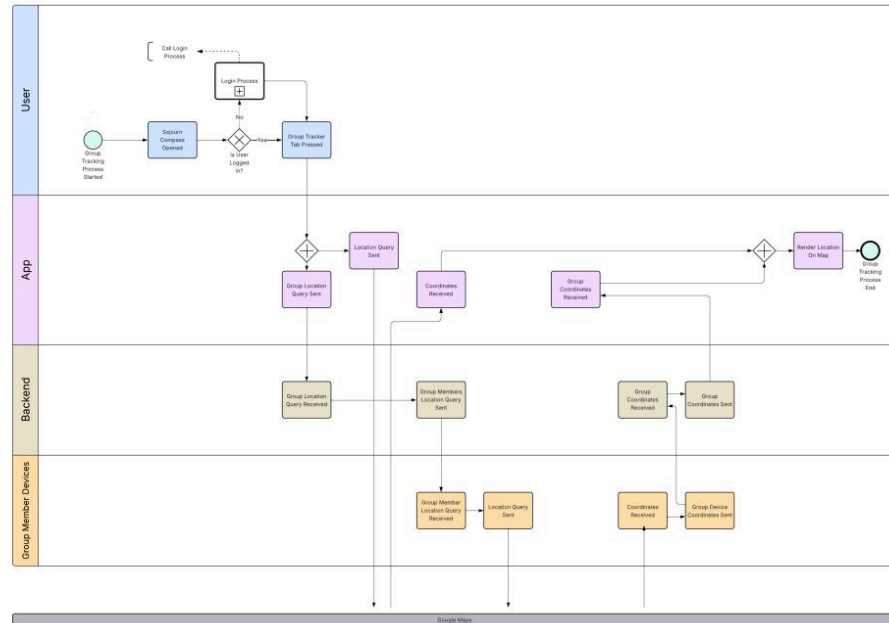


Figure 11: BPMN for Login Process

The Login Process begins when a user opens the Sojourn Compass application, triggering an immediate system check of existing authentication status. If the user is already logged in, the process bypasses credential entry and proceeds directly to displaying the Animo Compass home page. For unauthenticated users, the application prompts for credentials, which are securely transmitted to the backend server upon submission. The backend receives the credentials and performs verification against the users table, determining whether the provided information matches stored records. If credentials match, the server generates and sends an authentication token back to the application, where it is stored locally for session management before the home page is rendered. This token-based authentication ensures secure, persistent access throughout the pilgrimage while maintaining role-based permissions that distinguish between standard pilgrims and coordinators, thereby

protecting sensitive information and enabling personalized feature sets for each user type.

8.4. Group Tracking Process



The Group Tracking Process begins when a user opens the Sojourn Compass application and undergoes the standard login verification. If the user is not authenticated, the login process is invoked before proceeding; if already logged in, the user gains immediate access to the application interface. Upon pressing the Group Tracker tab, the system initiates the group tracking process, establishing communication with the backend server which coordinates location data across all group member devices. The backend aggregates real-time position information from each pilgrim's device and integrates this with Google Maps services to render a unified visual interface showing the geographical distribution of all team members. This process enables team leaders to conduct efficient digital headcounts, identify members who have wandered from designated meeting areas, and locate individuals requiring assistance, while standard pilgrims can view coordinator positions to navigate back to the group independently. The feature addresses the critical safety vulnerability of elderly participants becoming separated in unfamiliar European cities, transforming what was previously a

source of anxiety into a coordinated, technology-enabled assurance of group cohesion.

8.5. Daily Itinerary Process

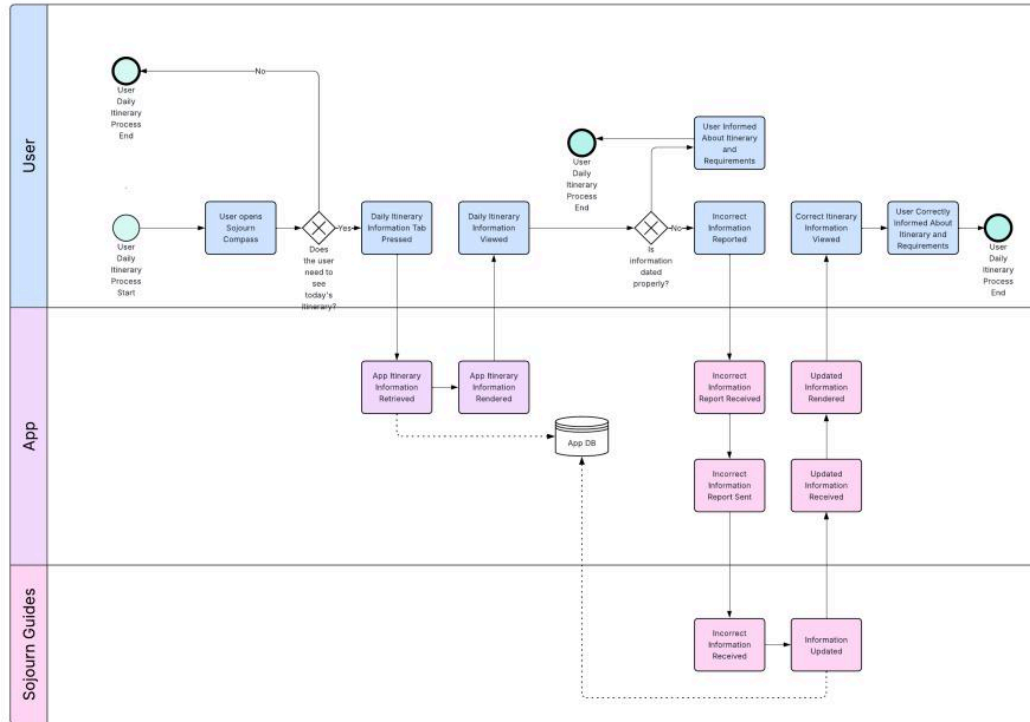


Figure 13: BPMN for Daily Itinerary Process

The Daily Itinerary Process begins when a user opens the Sojourn Compass application and determines whether they need to view the day's schedule. If the user selects the Daily Itinerary Information Tab, the application retrieves the relevant itinerary data from its local database and renders it on screen for immediate viewing. The user then evaluates whether the displayed information is current and accurate. If the information is properly dated, the process concludes with the user correctly informed about the day's itinerary and associated requirements. If the user identifies incorrect or outdated information, they can submit an incorrect information report directly through the application. This report is transmitted to the backend system, where solution guides are prepared and updated information is generated. The updated information is then pushed back to the application and rendered for the user, completing the corrective feedback loop. This process ensures that itinerary

information remains reliable despite the dynamic, frequently changing nature of pilgrimage schedules, directly addressing the documented problem of confusing Google sheets and verbally communicated updates that leave participants uncertain about daily activities and requirements.